

Stay Ahead of the Competition: An Approach for Churn Prediction by Leveraging Competitive Service App Usage Logs

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ABSTRACT

With the widespread adoption of smartphones, users now have easy access to similar services, leading to increased churn. As a result, it has become essential for service providers to prevent churn caused by customers' switch to competing services. The most common approach for service providers to prevent their customers' churn is to make churn predictions by monitoring customers' usage patterns of their own services. However, despite the importance of insights concerning customers' usage of competing services for the retention of customers, such information are yet to be integrated into churn prediction models due to the lack of suitable monitoring methods. Here, we propose an approach to predict user churn leveraging the event logs from smartphones and tablets. Instead of conventional churn prediction methods that solely rely on the users' usage patterns of their own service, our approach predicts churn by utilizing users' usage patterns of competing services, including their trial use of service before switch to competitor's. We evaluated the prototyped prediction model using smartphone logs collected from NTT DOCOMO smartphone and tablet users who consented to data collection between April 2020 and March 2021. The results demonstrated that the proposed method achieved AUC values ranging from 0.844 to 0.923. Moreover, our approach improved the performance of the conventional method that predicts churn without leveraging the features of the competitor's app by 1.8% to 7.5%.

CCS CONCEPTS

• **Information systems** → **Enterprise applications**; • **Computing methodologies** → *Classification and regression trees*.

KEYWORDS

Churn prediction; Mobile apps; Smartphone event log

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1 INTRODUCTION

Service providers need to retain their existing users in order to maintain their services. Moreover, with smartphones' rapid proliferation, people have plenty of similar services to choose from and can switch between them easily [4]. In this situation, it has become increasingly important for service providers to prevent user churn caused by their migration to similar services.

The most common approach for service providers to prevent churn is through prediction by monitoring user's usage of their service [7, 14, 16]. Mozer *et al.* [14] used a machine learning model to predict customer churn in a telephone company, using features such as the types of services the customer subscribed to, the date of subscription, monthly fees and usage, and the number of calls made.

On the other hand, surveying customers' usage of competitors' services helps providers prevent churn and retain customers at risk of proactively switching to competitors. Salo and Makkonen [15] found that users tend to switch smartphone services after trying out similar services offered by competitors, and they suggest that intervention by its providers during this trial phase can prevent users from leaving. Hence, monitoring users' usage of other companies' services can help service providers predict users' churn.

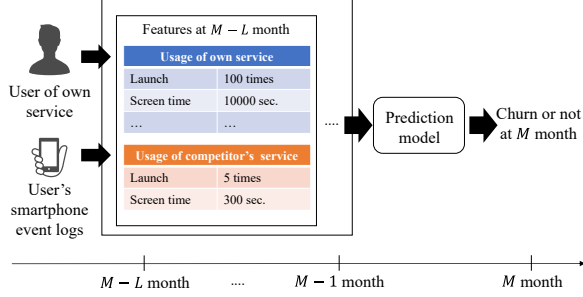
For service providers to survey their customers' usage of competing services, they can use one of three conventional methods known as competitive analysis:

- 1) Conducting online surveys with own customers [9]
- 2) Mining external sources such as SNS and social media with text mining techniques [5, 6, 8, 17]
- 3) Analyzing user traffic using tools such as Google Analytics [12] or Google Analytics for Firebase [13]

To predict customer churn based on the nuances observed in their usage of competitors' services, it is essential to survey customers' usage satisfying the following four criteria: (i) **user coverage** that covers sufficient number of users; (ii) **user segmentation** with sufficient segments user to analyze individual users; (iii) **data freshness** with regular data refresh for capturing up-to-date data churn dynamics; and (iv) **granularity of behavior** to obtain understanding on detailed usage patterns of competing services. Table

Table 1: Comparison of common competitive survey methods

Monitoring methods	Data freshness	User segmentation	User coverage	Granularity of behavior
Online surveys	Low	Individual	Narrow	High
Mining SNS and social media	High	Group	Narrow	Low
Analyzing user traffic	High	Group	Wide	Low


Figure 1: An overview of our proposed churn prediction method with user's smartphone logs

1 compares conventional monitoring methods based on those criteria.

Conducting online surveys: Not necessarily all customers respond to online surveys that are sent directly from their service providers, which results in low user coverage. Moreover, it may be difficult to conduct large-scale surveys frequently within a brief period because of the time and other burdens such surveys impose on customers, which could affect the freshness of the data collected.

Mining external sources: When service providers utilize text mining techniques to extract information from external sources such as SNS, social media, and review sites, they can gain insights into macro-level information, such as market trends. For instance, Dey *et al.* [5] conducted a study on sentiment analysis of social media to compare brand popularity for competing perfume brands. However, tracking individual usage of competitors' services is challenging as it requires linking external site accounts with their corresponding service accounts. Consequently, this method suffers from the limitations in churn prediction in terms of user coverage and granularity of behavior analysis.

Analyzing user traffic: When analyzing user traffic using tools such as Google Analytics and Google Analytics for Firebase, service providers can monitor user traffic in real-time to track customer transitions and access from external sites. However, this only provides aggregated results rather than detailed analysis at an individual level. Consequently, traffic analysis survey has limitations in conducting individual-level segmentation analysis and achieving granular behavior analysis for churn prediction.

Given the limitations of existing surveying methods that only partially meet the four criteria, they are not suitable for predicting customers' churn based on their usage of competitors' services.

Smartphone event logs provide real-time data and broad coverage as they track the apps opened and closed by the users. They also offer more granular behavioral analysis than tracking user traffic since it allows for surveying the usage of services from the point of

installation without depending on transitions between sites and applications. Therefore, surveying smartphone event logs is a superior approach to the conventional methods for surveying customers' usage of competitors' services, satisfying broad user coverage, user segmentation for individual surveillance, data freshness capturing up-to-date usage, and detailed granularity of behavior analysis.

In this paper, we propose an approach to predict user churn using a smartphone event log that provides valuable insights into users' usage of competitors' services. In addition, we prototype a machine-learning-based model that predicts churn by user features derived from event logs and evaluates it.

Figure 1 shows an overview of the proposed method. We propose a machine-learning-based method to predict churn, taking as input the customer's monthly usage patterns of the service provider's own app and competitive app within a recent period of L months. The prediction model determines whether the user will churn in the following month, denoted as M .

The contributions of this study are as follows:

Proposal of an approach and method: We proposed an approach for predicting user churn leveraging the usage of competing services obtained from users' smartphone event logs. To implement this approach, we prototyped a prediction model with LightGBM.

Evaluate the proposed approach: We evaluated our proposed method using smartphone logs obtained from NTT DOCOMO smartphones. The results demonstrated that the proposed method achieved AUC values ranging from 0.844 to 0.923 and improved the conventional method in AUC by a range between 1.80% and 7.51% across various apps in the "Entertainment" and "Music & Audio" categories. We also identified that the Top-10 effective features in each model included usage features related to the competitive apps. The result indicated that competitive app usage features enhanced the performance of churn prediction, and these features were more effective in the "Music & Audio" category than in the "Entertainment" category.

2 PROPOSED METHOD

As shown in Figure 1, we propose a machine-learning-based method for service providers to predict customer's churn based on their usage of competitor's apps. Throughout this study, we refer to the service provider's own app as A_{own} and its competitor's app as A_{comp} .

The motivation behind our proposed method is to enhance the performance of churn prediction by incorporating the usage patterns of A_{comp} as additional features to the conventional method that solely relies on the usage patterns of A_{own} . In particular, we aim to improve churn prediction performance by capturing signs of users switching from A_{own} to A_{comp} , such as users' trial of A_{comp} . For example, when SNS users switch to other service, they often

try alternative SNS before completely stopping using the previous one [15]. By detecting these early trial usages of A_{comp} using user's smartphone logs, we can improve the performance of churn prediction.

We denote the prediction model as f and $X_{i,M}$ as its input feature vector to provide a formal explanation. In addition, we denote user i 's features derived from usage of A_{own} and A_{comp} in t month as $x_{i,t}(A_{own})$ and $x_{i,t}(A_{comp})$, respectively. We describe the proposed model that predicts whether user i will churn in month M in the following formulation:

$$\begin{aligned} X_{i,M} &= (x_{i,M-L}(A_{own}), \dots, x_{i,M-1}(A_{own}), \\ &\quad x_{i,M-L}(A_{comp}), \dots, x_{i,M-1}(A_{comp})) \\ \widehat{y_{i,M}} &= \begin{cases} \text{Churn at } M & \text{if } 0.5 \leq f(X_{i,M}). \\ \text{Not churn at } M & \text{otherwise.} \end{cases} \end{aligned}$$

$\widehat{y_{i,M}}$ stands for prediction of user i 's churn at M .

For creating app usage features $x_{i,t}(A_{own})$ and $x_{i,t}(A_{comp})$ describing the user's usage of A_{own} and A_{comp} at month t , we can utilize several features capturing smartphone app usage, such as duration (screen time) and the number of launches.

Furthermore, we train the model with a dataset consisting of $(X_{i,t}, y_{i,t})$ pairs: $\mathcal{D} = \{(X_{i,t}, y_{i,t})\}_{1 \leq i \leq N}$. Here, N is the number of sampled users, t can be any month in the past, and $y_{i,t}$ is the ground truth regarding whether user i has churned at month t .

3 EVALUATION EXPERIMENT

We performed an evaluation experiment to validate the effectiveness of our approach using Android OS smartphone logs obtained from approximately 600,000 randomly selected users who consented to data collection agreement on NTT DOCOMO smartphones and tablets between April 2020 and March 2021.

To validate the effectiveness of our approach across various types of services, we first constructed multiple datasets $\mathcal{D}_M$ with various combinations of competing A_{own} and A_{comp} , and we trained and evaluated the proposed model comparing with conventional churn prediction approach. Detailed data creation instruction is provided in Section 3.1.

For creating datasets, we created multiple datasets $\mathcal{D}_M$:

$$\mathcal{D}_M = \{(X_{i,M}, y_{i,M})\}_{1 \leq i \leq N}$$

by varying the combinations of A_{own} and A_{comp} . Through the evaluation, we set M as October 2020 and L as six months. Therefore, we created $\mathcal{D}_M$ based on whether users churned in October 2020, and we trained and evaluated models to predict churn in October 2020.

For comparing the proposed model with conventional churn prediction approach, we trained two prediction models for each $\mathcal{D}_M$: **Baseline method** that utilizes the app usage features of A_{own} , and **Proposed method** that incorporates both the app usage features of A_{own} and A_{comp} . Through the evaluation, we exploited LightGBMs [1, 10] for the baseline and proposed methods setting M as October 2020 and L as six months. In other words, we trained and evaluated LightGBMs that predict whether the users will churn in October 2020 based on the monthly app usage of A_{own} and A_{comp} from April 2020 to September 2020. Detailed instruction for data creation is provided in the following section.

3.1 Creating datasets

To create $\mathcal{D}_M$ with various combinations of competing apps, we first selected multiple combinations of two competitive Android apps as A_{own} and A_{comp} . And then, for each combination, we converted the user's usage of A_{own} and A_{comp} from the collected event logs into $X_{i,M}$ and attached churn labels $y_{i,M}$ to each $X_{i,M}$.

Selecting combinations A_{own} and A_{comp} : For generating various combinations of A_{own} and A_{comp} , a two-step process was followed. Firstly, the Top-10 apps with the highest cumulative download numbers were selected from the "Music & Audio" and "Entertainment" categories [2, 3]. Secondly, from these selected ten apps, we identified two apps with the highest number of users who launched the apps in April 2020 using the collected event logs. Hereinafter, we use the notation A_{m-1st} and A_{m-2nd} to refer to the most popular app and second most popular app in the "Music & Audio" category. Likewise, for the "Entertainment" category, we use A_{e-1st} and A_{e-2nd} .

As a result, we designed four experimental setups for creating datasets: $\mathcal{D}_{\sim A_{m-1st}}$, $\mathcal{D}_{\sim A_{m-2nd}}$, $\mathcal{D}_{\sim A_{e-1st}}$, and $\mathcal{D}_{\sim A_{e-2nd}}$.

Note that $\mathcal{D}_{\sim A_{m-1st}}$ represents the dataset created considering A_{m-1st} as A_{own} and A_{m-2nd} as A_{comp} . The same notation convention applies to the other datasets. In the following sections, $\mathcal{D}_M$ refers to each of these four datasets.

Converting the event logs into $X_{i,M}$: For converting the event logs into features $X_{i,M}$, we exploited two features for A_{own} and A_{comp} : the total duration time (screen time) and the number of app launches per month; we created 12-dimensional feature for the input of the baseline method, and we created 24-dimensional feature for the input of the proposed method. Note that, Android OS event log captures the user's app operations with timestamps as events, including app transitions to the foreground or background [11]. Therefore, we determined the number of app launches within a month by tallying the number of transitions to the foreground. Similarly, we calculated the screen time within a month by summing the time elapsed from foreground transition to background transition.

Attaching churn labels $y_{i,M}$: To attach churn labels $y_{i,M}$ to each $X_{i,M}$, we determined the churn status $y_{i,M}$ of each user by identifying those who did not launch A_{own} continuously in the period between October 2020 and March 2021.

Extracting negative and positive samples: To create the dataset $\mathcal{D}_M$ for training and evaluation, we employed the following approach for sampling negative and positive examples: For negative sampling, we selected users who used both A_{own} and A_{comp} between April 2020 and September 2020 but did not churn from A_{own} in October 2020. For positive sampling, we selected users who used both A_{own} and A_{comp} between April 2020 and September 2020 and churned from A_{own} in October 2020 but continued using A_{comp} .

Furthermore, downsampling was applied to match the number of positive samples for the negative samples.

As a result, we obtained four datasets: $\mathcal{D}_{\sim A_{m-1st}}$ comprising 2126 users, $\mathcal{D}_{\sim A_{m-2nd}}$ comprising 1548 users, $\mathcal{D}_{\sim A_{e-1st}}$ comprising 1772 users, and $\mathcal{D}_{\sim A_{e-2nd}}$ comprising 2540 users.

Table 2: The mean values of the evaluation results on the four datasets

	Evaluation on $\mathcal{D}_{\sim A_{m-1st}}$			Evaluation on $\mathcal{D}_{\sim A_{m-2nd}}$			Evaluation on $\mathcal{D}_{\sim A_{e-1st}}$			Evaluation on $\mathcal{D}_{\sim A_{e-2nd}}$		
	AUC	Precision	Recall	AUC	Precision	Recall	AUC	Precision	Recall	AUC	Precision	Recall
Baseline model	0.785	0.702	0.856	0.877	0.792	0.888	0.856	0.760	0.843	0.907	0.814	0.863
Proposed model	0.844*	0.739	0.823	0.915*	0.818	0.903	0.875	0.783	0.866	0.923*	0.840	0.875
Improvement in AUC (%)	7.51%			4.32%			2.14%			1.80%		

Table 3: Top-10 contributed features for each proposed model

Proposed method on $\mathcal{D}_{\sim A_{m-1st}}$	Proposed method on $\mathcal{D}_{\sim A_{m-2nd}}$	Proposed method on $\mathcal{D}_{\sim A_{e-1st}}$	Proposed method on $\mathcal{D}_{\sim A_{e-2nd}}$
Screen time of A_{own} in 2020/9	Launches of A_{own} in 2020/9	Screen time of A_{own} in 2020/9	Screen time of A_{own} in 2020/9
Launches of A_{own} in 2020/9	Screen time of A_{own} in 2020/9	Launches of A_{own} in 2020/9	Screen time of A_{own} in 2020/7
Screen time of A_{comp} in 2020/9	Launches of A_{comp} in 2020/9	Screen time of A_{own} in 2020/7	Screen time of A_{own} in 2020/4
Screen time of A_{own} in 2020/8	Screen time of A_{own} in 2020/8	Screen time of A_{own} in 2020/8	Screen time of A_{comp} in 2020/9
Launches of A_{comp} in 2020/9	Screen time of A_{own} in 2020/7	Screen time of A_{own} in 2020/6	Screen time of A_{own} in 2020/8
Launches of A_{comp} in 2020/7	Screen time of A_{comp} in 2020/8	Screen time of A_{own} in 2020/4	Screen time of A_{own} in 2020/6
Screen time of A_{comp} in 2020/7	Screen time of A_{comp} in 2020/9	Screen time of A_{comp} in 2020/8	Launches of A_{own} in 2020/9
Screen time of A_{comp} in 2020/8	Screen time of A_{own} in 2020/5	Screen time of A_{comp} in 2020/9	Screen time of A_{own} in 2020/5
Screen time of A_{comp} in 2020/6	Launches of A_{own} in 2020/7	Launches of A_{own} in 2020/8	Screen time of A_{comp} in 2020/7
Screen time of A_{own} in 2020/7	Screen time of A_{own} in 2020/6	Launches of A_{own} in 2020/7	Screen time of A_{comp} in 2020/8

3.2 Results and Discussion

To train and evaluate both the baseline and proposed methods, we conducted five-fold cross-validation for each dataset. Table 2 presents the performance of the baseline and proposed methods on each dataset. The first row indicates the datasets used for evaluation. Each column represents the evaluation results for the AUC, precision, and recall metrics, comparing the performance of the baseline and the proposed methods. The values represent the mean values of the evaluation results obtained through cross-validation. The AUC value with * indicates the performance with a significant difference between the two models confirmed by the Brunner-Munzel test ($p < 0.05$).

As shown in Table 2, the proposed method achieved performance ranging from a minimum precision value of 0.702 to a maximum of 0.840 and recall values ranging from 0.823 to 0.903. These results indicate that the proposed approach can predict churn across various app categories.

In addition, the proposed method outperformed the baseline method with respect to AUC across all datasets. These findings suggest that incorporating customer's usage features of competitive apps improves the performance of churn prediction.

Moreover, we assessed the feature importance by calculating the average gain values of LightGBMs during the five-fold cross-validation process. The top-contributed features for each proposed model are shown in Table 3. The Top-10 features in all proposed models included features related to A_{comp} . This finding suggests that incorporating competitive app usage features enhances the performance of churn prediction. However, when comparing the Top-10 features between the "Music & Audio" and "Entertainment" categories, the features related to A_{comp} ranked lower in the churn prediction model for the "Entertainment" category. Therefore, the impact of competitive app usage on churn prediction performance varies depending on the app category.

To further analyze the effectiveness of features related to A_{own} across different app categories, we calculated the improvement rate of the proposed method over the baseline method in terms of AUC for each dataset. The results are presented in the bottom row of Table 2, which show that the improvement rate in the "Music & Audio" category was higher than in the "Entertainment" category.

These findings indicate that the effectiveness of the proposed method's improvement varies depending on service categories. This discrepancy in improvement rates could be attributed to the varying levels of competition among different service categories. For example, in the "Music & Audio" category, users typically use only one music streaming app at a time, whereas in the "Entertainment" category, users tend to use multiple competing movie streaming apps simultaneously.

Furthermore, we calculated the average period in which users used both A_{own} and A_{comp} in the same month from April 2020 to September 2020 for each dataset. As a result, even among users who did not churn in October 2020, the average duration of using both apps in the same month was 1.6 months in $\mathcal{D}_{\sim A_{m-1st}}$, 1.8 months in $\mathcal{D}_{\sim A_{m-2nd}}$, 2.8 months in $\mathcal{D}_{\sim A_{e-1st}}$, and 3.0 months in $\mathcal{D}_{\sim A_{e-2nd}}$. These differences in average durations also indicate that even among users who did not churn, the "Entertainment" category apps tend to be used together with other entertainment apps more frequently than the "Music & Audio" category apps.

4 CONCLUSION

In this paper, we presented an approach for predicting churn by leveraging users' usage features of competitor services extracted from smartphone event logs. We developed a machine-learning-based prediction model that incorporates the usage features of the service provider's own app and its competitor's app. Through our evaluation, we demonstrated the effectiveness of our proposed approach. While the findings are still preliminary, our research

makes valuable contributions to the field of churn prediction. Our approach offers insights for service providers seeking to enhance their retention strategies taking into consideration customers' engagement with competing services. We hope our work inspires further research and advancements in churn prediction techniques.

REFERENCES

- [1] Abdelrahim Kasem Ahmad, Assef Jafar, and Kadan Aljoumaa. 2019. Customer churn prediction in telecom using machine learning in big data platform. *Journal of Big Data* 6, 1 (2019), 1–24.
- [2] AppBrain. (Accessed: 2 June 2023). The 20 most popular Free Entertainment Apps for Android ever. <https://www.appbrain.com/apps/popular/entertainment/free>
- [3] AppBrain. (Accessed: 2 June 2023). The 20 most popular Free Music & Audio Apps for Android ever. <https://www.appbrain.com/apps/popular/music-and-audio/free>
- [4] Juliette Caminade and Markus von Wartburg. 2022. The Success of Third-Party Apps on the App Store. <https://www.apple.com/newsroom/pdfs/the-success-of-third-party-apps-on-the-app-store.pdf>
- [5] Lipika Dey, Sk Mirajul Haque, Arpit Khurdiya, and Gautam Shroff. 2011. Acquiring competitive intelligence from social media. In *Proceedings of the 2011 joint workshop on multilingual OCR and analytics for noisy unstructured text data*. 1–9.
- [6] Wu He, Xin Tian, Yong Chen, and Dazhi Chong. 2016. Actionable social media competitive analytics for understanding customer experiences. *Journal of Computer Information Systems* 56, 2 (2016), 145–155.
- [7] Bingquan Huang, Mohand Tahar Kechadi, and Brian Buckley. 2012. Customer churn prediction in telecommunications. *Expert Systems with Applications* 39, 1 (2012), 1414–1425.
- [8] Jian Jin, Ping Ji, and Rui Gu. 2016. Identifying comparative customer requirements from product online reviews for competitor analysis. *Engineering Applications of Artificial Intelligence* 49 (2016), 61–73.
- [9] Douglas M. Lambert and Arun Sharma. 1990. A customer-based competitive analysis for logistics decisions. *International Journal of Physical Distribution & Logistics Management* 20, 1 (1990), 17–24.
- [10] Aurélie Lemmens and Christophe Croux. 2006. Bagging and boosting classification trees to predict churn. *Journal of Marketing Research* 43, 2 (2006), 276–286.
- [11] Google LLC. (Accessed: 2 June 2023). Android Developers Documentation. <https://developer.android.com/reference/android/app/usage/UsageEvents.Event>
- [12] Google LLC. (Accessed: 2 June 2023). Google Analytics. <https://analytics.google.com/>
- [13] Google LLC. (Accessed: 2 June 2023). Google Analytics for Firebase. <https://firebase.google.com>
- [14] Michael C. Mozer, Richard Wolniewicz, David B. Grimes, Eric Johnson, and Howard Kaushansky. 2000. Predicting subscriber dissatisfaction and improving retention in the wireless telecommunications industry. *IEEE Transactions on neural networks* 11, 3 (2000), 690–696.
- [15] Markus Salo and Markus Makkonen. 2018. Why do users switch mobile applications?: Trialing behavior as a predecessor of switching behavior. *Communications of the Association for Information Systems* 42 (2018). Publisher: Association for Information Systems.
- [16] Thanasis Vafeiadis, Konstantinos I. Diamantaras, George Sarigiannidis, and K. Ch Chatzisavvas. 2015. A comparison of machine learning techniques for customer churn prediction. *Simulation Modelling Practice and Theory* 55 (2015), 1–9.
- [17] Kaiquan Xu, Stephen Shaoyi Liao, Jiexun Li, and Yuxia Song. 2011. Mining comparative opinions from customer reviews for competitive intelligence. *Decision support systems* 50, 4 (2011), 743–754.